

Visiting the 61" telescope on Mt Bigelow

Once all the observing groups have chosen a topic for their observing project, each group will be assigned a specific one of the scheduled ASTR302 observing nights to go up to the 61" (1.55m) telescope on Mt. Bigelow. You will get an introduction to the telescope and observing procedures and, depending on your project, a chance to observe your target(s) with the Mont4k CCD camera, weather permitting. Your group will be driven up to the 61" by your TA in a UA minivan, leaving campus in the afternoon and returning either late in the evening or the following morning (see below), depending on the weather and whether your target can be observed in the first half of the night or if it needs to be observed later in the night. If your group is unable to get all the observations needed for your project on your scheduled telescope night, we will try to reschedule your observations to an ASTR302 night later in the semester if/when the weather makes it possible to try again. Usually the 61" observer (Green) will obtain the subsequent observations for you, while you watch the progress of the observations using Zoom. Groups working with archival or data from another telescope typically also go up to the 61" for a similar but shorter introduction to the telescope, and they will return relatively early in the evening.

This semester ASTR302 has been allocated 61" observing time on four weekends from mid-February to the mid- March. We try to take observing groups up to the telescope on Friday or Saturday nights, or during the first half of the night on Sunday nights, so you won't miss classes on Monday morning. **This class is scheduled to observe during the weekends of Feb 20-22, Mar 6-8, Mar 13-15 and Mar 20-22. Each group should try to do their observing as early in the semester as possible, in order to have more options for make-up nights if your first assigned night is weathered out.** Note that many targets can only be observed on specific nights: exoplanet transits and eclipses can only be observed on a night when those events are occurring, and faint stars, galaxies or extragalactic supernovae can only be observed when the sky is dark enough, ie within several days of the new moon or when the moon is below the horizon. We will help you choose 61" telescope nights that will be suitable for your group to observe, depending on your target and your schedule constraints and the needs of the other groups.

Important Covid precautions

For any trip to the telescope, you will be spending time in very close quarters with your fellow students, TA and instructor during the 70 minute drive in the UA vehicle on the way to and from the telescope, and for many hours in the telescope control rooms, which is very small and not well ventilated.

Because some people are at high risk, everyone should plan to take a Covid test on the day that they go up to a telescope, *before* they meet their instructor or TA to be driven up to the mountain in the minivan. If you can't take a Covid test the day of your telescope visit or you test positive, you will need to postpone your trip to the mountain and go up with a later group. Everyone should also bring a good masks with them (eg N95, KF94 or similar) in case they are needed.

61" Logistics

On the day you go up to the 61" telescope to observe, the UA minivan will be parked in the loading zone on Cherry Avenue outside the main lobby / front door of Steward, by about 2:10 pm in the afternoon. The van will leave for the telescope at 2:20 pm. Please be on time!!

Those of you who are scheduled to go up to the 61" on a Friday may drop off your food and gear in the main astronomy office (the room with glass windows next to the main lobby) anytime in the morning and pick it up at 2 pm before you meet the TA and the minivan. The office staff will keep your things safe for you in the back room, so you don't have to lug everything around with you to your earlier classes.

It takes about 70 – 75 minutes to drive up to the 61" telescope, so it's recommended to visit the restroom in the Steward lobby before you go.

You will need to bring:

- 1) a screenshot of your negative Covid test result on your phone for your instructor to verify before you get in the van,
- 2) a good mask (N95, KF94 or similar),
- 3) food and drink for dinner that evening (sandwiches, leftovers, takeout, anything that can be eaten cold or easily heated/reheated in a microwave), plus enough snacks to get you through the night and whatever breakfast you'll need the next morning, or until your TA drives you back to campus. The 61" control room has a small kitchen with a microwave, an old coffee maker, a toaster, a few stove burners, and a tiny oven; also a few pots and pans, silverware, plates, bowls, cups. There are usually a few items left from previous observers, but you should bring all food and drink (besides water) that is important to you, including your own caffeine. The tap water on the mountain is fine to drink; it comes from a spring on top of Mt Lemmon,
- 4) a sweatshirt or jacket; the dome temperatures could easily be in the mid-40's or colder and we'll be spending some time in the dome,
- 5) a pen or pencil for taking notes,
- 6) if you will observe past midnight, your TA may not want to drive down the mountain that late, and the dorm supplies only blankets and pillows. so you will need to bring a sleeping bag or sheets, a pillowcase, and a hand towel (optionally you may also prefer to bring your own pillow),
- 7) a small flashlight if you have one; phone flashlights run down your battery rather fast; also phone flashlights are too bright to use in the dome while observations are being taken.

If you have any questions, email Betsy Green (egreen@arizona.edu).